Pullela Giridhar

Portfolio | LinkedIn | GitHub | giridharpullela2003@gmail.com | +91 75693 45747
Plot L2&L3, Serene Noor, Road No.10/B, Panchavati Colony, Manikonda, Hyderabad – 500089

Education

B.Tech, CSE-Cybersecurity & Blockchain Technology | *SASTRA Deemed University, Tamil Nadu* — Oct 2022 – May 2026

- **CGPA: 7.244/10**

Publications

- P. Lavanya, P. Vaishnavi, **P. Giridhar**, H. Anila Glory, V. S. Shankar Sriram. "Scenario Driven Insider Threat Detection Using Dual Modelling Architecture." *LNNS vol. 1938, pp. 129–143.* Springer Nature, 2026. **Published.**
  DOI: 10.1007/978-3-032-23945-7_11
- P. Vaishnavi, V. Senthilkumar, **P. Giridhar**, D. S. Pujitha, D. Nithyakalyani, V. Thirugnanasambandam. "Generative AI-Driven Self-Healing and Adaptive Spectrum Management for Resilient Satellite Communications." *IEEE SPACE 2026.* **Accepted.**
- Lavanya P, Vaishnavi P, **Giridhar P**, et al. "CIGNet-XIIoT: Cluster-Integrated Graph-Enhanced Framework for Insider Threat Detection in IIoT Networks." *Target: IEEE Trans. on Network & Service Management.* **In Preparation.**

Research Experience

Research Intern, LLM Security | *Colorado State University* | *Supervisor: Prof. Indrakshi Ray* — Jan 2026 – Present

- Developing a secure RAG chatbot (AMI), benchmarking centralized, kernelized, and distributed RAG architectures under adversarial attack scenarios
- Implementing NVIDIA NeMo Guardrails for prompt-injection defense and LLM output safety enforcement across all three pipeline configurations

Research Lead, CIGNet-XIIoT | *SASTRA University* | *IIoT Insider Threat Detection* — Jul 2025 – Present

- Designed a four-stage pipeline (RIPT-XIIoT → TriScore-IGX → GEKME-X → GCPT-X) on 820K+ IIoT samples; achieved 98.14% classification accuracy
- Proposed a novel BCR loss and Cluster-Initialized Prototype Attention (CIPA) bridging unsupervised clustering with supervised classification

Research Contributor, Self-Healing Satellite | *SASTRA University* | *IEEE SPACE 2026* — Mar – Dec 2025

- AI pipeline: TimeGAN (spectrum forecasting), CNN + Vision Transformer (99.70% damage classification, 9 classes), Edge-Connect GAN for structural repair (SSIM 0.92)
- Integrated GPT-2 LLMs for real-time operator decision support; copyright in progress

Research Lead, DMA-PULSE | *SASTRA University* | *Scenario-Driven Insider Threat Detection* — Feb – Aug 2025

- Self-Attention Transformer with AOA–RFO metaheuristic optimizer; 97.67% multi-class accuracy, 0.012 FPR
- Designed the PULSE Engine: a synthetic user-activity simulator generating 14,010 behavioral sequences for binary and multi-class insider threat classification

Research Lead, FUMVar-Ex | *SASTRA University* | *Adversarial Malware Mutation Framework* — Jan – May 2025

- Malware mutation pipeline (Genetic Algorithms) generating obfuscated samples that evade ML-based antimalware; used Cuckoo Sandbox and Hybrid Analysis for dynamic testing, MalwareBazaar API for sample collection
- Automated mutation decisions using Gemini CLI via an MCP server based on sandbox outputs; all experiments run in isolated Docker containers

Industry Experience

GenAI Intern | *Mitisphere Solutions, Hyderabad* — May – Jun 2024

- Built a LangChain LLM summarization pipeline for enterprise tender documents; developed Google Calendar and MS Teams API integrations, improving workflow efficiency by 45%

Technical Skills

Languages	Python, Bash, C, JavaScript
ML	PyTorch, TensorFlow, Scikit-learn
Security Tools	Cuckoo Sandbox, Hybrid Analysis, MalwareBazaar API, Burp Suite, Wireshark
AI & LLM	NeMo Guardrails, LangChain, RAG Architectures, MCP Server, Gemini CLI
Dev & Infra	Docker, Git/GitHub, Linux/Unix

Certifications, Leadership & Achievements

- **CompTIA Security+ (SY0-701) | Certificate ID: COMP001022849282** — Aug 2026
- **Cyber Security Certification**, Edureka Institute | Certificate ID: PSNDVC19J — Aug – Oct 2023
- **Research Team Lead**: Led 3-member teams across five AI security research projects
- **Cybersecurity Batch Mentor**, SASTRA : Mentored 60+ students; co-authored a departmental lab manual adopted institution-wide
- **Best Paper Award:** “Generative AI-Driven Self-Healing and Adaptive Spectrum Management for Resilient Satellite Communications,” IEEE SPACE 2026